



RUNBOOK

VIRTUAL CONTROL

VMWARE CLOUD FOUNDATION SOLUTIONS

VCF 9.0.1 → 9.0.2 Patch Precheck Remediation

Clearing a 26-error / 4-warning patch precheck down to zero. The errors collapse to four root causes — unstaged depot bundle, a missing vCenter service account, unconfigured NSX backup, and a nested-hardware vSAN controller check. Each with its verified fix and the “don’t-do-this” caveats.

v1.1

VMware Cloud Foundation 9.0.1 → 9.0.2

VCF 9.0.1 → 9.0.2 Patch Precheck Remediation Runbook

Prepared by: Virtual Control LLC **Date:** July 29, 2026 **Environment:** VMware Cloud Foundation 9.0.1 (mgmt domain, nested homelab) **Target:** 9.0.2.0 patch — SDDC Manager, vCenter, NSX, ESX **Precheck result being remediated:** 26 errors, 4 warnings (completed 2026-07-29 20:00:59)

The whole point in one line: the **26 errors collapse to ~4 real root causes** — the count is high only because the same checks repeat per component. Fix the four, re-run, drive to **zero errors** (Broadcom treats the precheck as a gate; a failed precheck *may cause the update to fail*; **warnings do not block**).

Verification legend (this runbook, v1.1 — verified against Broadcom TechDocs/KB 2026-07-29): [V] = confirmed by a cited Broadcom source · [V-LIVE] = confirmed pattern, but verify the specific value in *your* UI/appliance before acting · [UNVERIFIED] = could not be confirmed from an authoritative source — do not treat as fact.

Where the precheck lives (VCF 9.0): VCF Operations → **Fleet Management** → **Lifecycle** → **VCF Instances** → [domain] → **Updates** → **Precheck** (the standalone SDDC Manager LCM path is retired in 9.0). Silence/Restore and the ALL/ERRORS/WARNINGS/SILENCED filters carry over. [V]

1. The Precheck Result (verified 2026-07-29)

Errors (26) — grouped by the actual failing check:

Component	Failing check	Root cause #
ESX Host Cluster domain-c10	Upgrade bundle download status	1
ESX Host Cluster domain-c10	vCenter patch store URL validation	1
ESX Host Cluster domain-c10	ESX Host checks	1 (depot-dependent) / re-run
ESX Host Cluster domain-c10	vSAN SCSI controller	4
NSX nsx-vip.lab.local	Upgrade bundle download status	1
NSX nsx-vip.lab.local	NSX backup	3
vCenter vcenter.lab.local	Upgrade bundle download status	1
vCenter vcenter.lab.local	Check Service Account	2

Component	Failing check	Root cause #
SDDC Manager <code>sddc-manager.lab.local</code>	Upgrade bundle download status	1

Warnings (4) — NON-blocking, do not gate the patch:

- ESX Host Cluster `domain-c10` — **Cluster DRS**
- NSX — **NSX cluster DRS**
- NSX — **NSX certificate CN**
- NSX — **NSX cluster backup configuration schedule** (*pairs with root cause #3*)

Current versions (all 9.0.1): SDDC Manager `9.0.1.0.24962180` · vCenter `9.0.1.0.24957454` · ESX `9.0.1.0.24957456` (4 hosts) · NSX `9.0.1.0.24952111` .

2. Root Cause #1 (DOMINANT) — 9.0.2 bundle not staged in the offline depot

"Upgrade bundle download status" fails on ALL FOUR components, plus **"vCenter patch store URL validation."** That is **one** problem, not five: the **9.0.2 bundle is not present in the offline depot** (`https://192.168.1.160:8443`). Every component checks "do I have my bundle?" and the answer is no. **Confirmed 2026-07-29: bundle "not yet" downloaded** — so this is the bulk of the 26 errors.

- **"Upgrade bundle download status" [V-LIVE]** — Broadcom depot/bundle docs confirm bundles stage per-component and version-specific prechecks only run once a bundle is downloaded; the *literal check-name definition* is **[UNVERIFIED]** (no page names that exact string), but the function is confirmed.
- **"vCenter patch store URL validation" [V]** — validates vCenter's software-depot/patch-store URL is reachable and returns a valid depot index. Common failures: dead legacy `vmware.com` URLs (now token-based `d1.broadcom.com`), or a proxy misconfig (LCM needs `HTTPS_PROXY` set). (KB 393951; see also KB 390120.)

Fix: 1. Confirm the **offline depot** is reachable and configured: base URL `https://192.168.1.160:8443` (NO `/PROD` suffix — the fleet appends it), locker alias/creds `admin / admin` , and a placeholder Customer Connect username entered (empty is rejected). (See *[[reference_offline_depot_config_values]]* and *[[reference_vcf_depot_server]]* — depot server is `I:\VCF-Depot\https_server.py` .) 2. **Download / stage the 9.0.2 bundle** into the depot (VCF Download Tool for air-gapped), so all four components' bundles are present. Confirm the depot shows **Connected** and the Broadcom **download token** is valid (VCF Operations Fleet → Binary/Depot management). **[V]** 3. **Re-run the precheck**. Expect the four "Upgrade bundle download status" errors, the "vCenter patch store URL validation" error, and the depot-dependent "ESX Host checks" to clear.

You cannot patch to a version whose bundle you do not hold. If the bundle is not staged, stop here and stage it first — everything else is secondary.

3. Root Cause #2 — vCenter service account missing (vCenter-redeploy fallout)

vCenter → **"Check Service Account"** fails. [V] Broadcom (KB 399460 / 314177) confirms this validates the VCF-managed vCenter service account `svc-<vcenter-fqdn>@<sso-domain>` — that it **exists, is not expired/locked, and its password matches** what VCF has stored. Documented causes: password drift between VCF and vCenter, **90-day SSO service-account expiry**, or the account **missing/locked** after an out-of-band vCenter redeploy. Corroborating tell here: the **Available Configuration Updates** list offers **"Create Service Account in vCenter"** — suggesting this account is **missing**, not merely drifted.

⚠ **DECIDE MISSING vs. DRIFTED FIRST — the two fixes are different, and the wrong one causes harm.** - **Broadcom's documented path for a DRIFTED/expired/locked account** is: reset/unlock `svc-<vcenter-fqdn>@<sso-domain>` in vCenter SSO, then **Remediate** in VCF Operations → Fleet → password management (KB 399460, KB 314177). [V] - **But if the account is MISSING**, do **NOT** use Remediate — this lab's own history shows Remediate on a *missing* service account **jams resource locks** (see `[[reference_sddc_nsx_service_account_missing]]`). For a missing account, **recreate** it (apply the **"Create Service Account in vCenter"** config update, or the `dir-cli` method KB 402863 / 327195). - **How to tell:** check whether `svc-<vcenter-fqdn>@<sso-domain>` **exists** in vCenter SSO users. Exists but stale → Remediate. Absent → recreate. [V-LIVE]

Fix — path A (account MISSING, likely here): apply the **"Create Service Account in vCenter"** configuration update from Available Configuration Updates, or recreate via `dir-cli` — see `[[reference_sddc_vcenter_service_account_recreate]]` and `[[MODULE-SDDC-Manager-Credential-Reconnect-After-vCenter-Redeploy-2026-06-13]]`.

Fix — path B (account EXISTS but drifted/expired/locked): reset/unlock in vCenter SSO → **Remediate** in VCF Operations password management (KB 399460 / 314177).

Re-run the precheck; **"Check Service Account"** should clear.

4. Root Cause #3 — NSX backup not configured

NSX → **"NSX backup"** error, with the paired warning **"NSX cluster backup configuration schedule."** [V] Broadcom (KB 330382) gives the exact check: *"Check if backup is configured and at least one backup is available."* So it requires a **configured NSX Manager backup with ≥ 1 successful backup present**.

Documented most-common cause [V]: the **SDDC backup user password on SDDC Manager has expired**, so scheduled NSX backups silently stop succeeding → no available backup. (Other causes: backup never configured, or the SFTP target is unreachable.)

Fix — step 1, rule out the expired backup user (KB 330382) [V]:

```
ssh vcf@sddc-manager → su -
chage -l backup      # check expiry
passwd backup        # reset if expired
chage -l backup      # verify
```

Then trigger an **on-demand NSX backup** so "at least one backup is available," and re-run the precheck.

Fix — step 2, if backup was never configured: set up NSX Manager file-based backup + schedule (NSX UI → System → **Backup & Restore**). **[V-LIVE]** (confirm the menu path in your NSX version) - SFTP target: the `.160` server; **use RELATIVE chroot paths** (e.g. `/nsx-manager`), never `C:/...` — see `[[reference_sftp_backup_chroot_path]]`. - **Fingerprint format matters:** NSX-T accepts **only ECDSA SHA256**; it rejects RSA/ED25519 variants — see `[[reference_nsx_sftp_fingerprint_format]]`.

Re-run the precheck; **"NSX backup"** error clears and the schedule warning resolves.

5. Root Cause #4 — vSAN SCSI controller (nested-hardware limitation)

ESX → "vSAN SCSI controller" — [V] confirmed via **KB 400362**: this check "checks if the vSAN disk group type (All-Flash or Hybrid) is certified for the used SCSI controller in the VMware Compatibility Guide" — i.e. a pure **HCL / hardware-compatibility** check (not a driver/firmware or controller-mode check). On this lab's **nested hardware, virtual/PVSCSI controllers are not on the vSAN HCL, so it will always fail** — a guaranteed false positive.

KB 400362 scope note [V]: "This check applies only to vSAN OSA clusters. Issues reported for other cluster types can be disregarded." → If your cluster is **ESA**, Broadcom says disregard it outright. This lab has **OSA** history, so the check likely does apply — but it's still a false positive on nested HW.

Fix (supported on 9.0): UI "Silence Precheck". After the depot + service-account + backup fixes and a re-run, if this is still red, **silence it** in the VCF Operations precheck UI (the domain already has **1 precheck silenced**, so the mechanism is in use). **[V]** Broadcom's guardrail (verbatim): "Only silence alerts if you know that they are incorrect. Do not silence alerts for real issues that require remediation." A known nested-HW false

positive qualifies; on production HCL hardware it would be a real remediation, not a silence. **RESTORE PRECHECK** reverses it.

⚠ **Do NOT use the backend property-edit bypass on 9.0.** KB 386879 ("Bypass SDDC Manager pre-check for vSAN Storage Controller" — editing `application-prod.properties` `vsan.precheck.enabled=false` + `systemctl restart lcm`) is **scoped to VCF 4.x/5.x** and assumes a standalone SDDC Manager LCM service. Its applicability to VCF 9.0's Fleet/VCF-Operations lifecycle is **[UNVERIFIED]** — the file/service path may not exist. **Use the supported UI Silence Precheck on 9.0;** do not blind-edit properties.

6. Non-Blocking Warnings — note, don't chase

Warning	Disposition
Cluster DRS (ESX cluster)	[V] Broadcom (KB 426719) wants DRS Enabled + Fully Automated with N-1 host capacity so hosts can auto-enter maintenance mode and vMotion VMs during the rolling upgrade. This is the DRS <i>automation level</i> — separate from the <i>VmotionRate threshold</i> (Conservative). Confirm automation = Fully Automated; non-blocking.
NSX cluster DRS	Same DRS-Fully-Automated requirement, evaluated for the cluster hosting NSX components during the NSX upgrade phase. (<i>Whether this is a truly distinct named check vs. the above is [UNVERIFIED]; same fix.</i>) Non-blocking.
NSX certificate CN	[V] Cert CN/SAN should be the FQDN, non-wildcard , matching the node/VIP (KB 419000). On 9.0.1 this is likely a real cert-hygiene item (the pre-9.0.1 wildcard-import false-positive is already fixed) — address at a cert-rotation window, not for this patch. Non-blocking.
NSX cluster backup configuration schedule	[V] Resolves with the NSX backup fix (§4). Note (KB 428140): pre-9.0.2, SDDC Manager auto-re-enables a disabled NSX backup schedule every ~5 min — fixed in 9.0.2. Non-blocking.

7. Remediation Order & The Gate

1. **Stage the 9.0.2 bundle in the depot** → re-run precheck (*clears the ~5 depot/bundle errors — the majority*).
2. **Apply "Create Service Account in vCenter"** → re-run (*clears the vCenter service-account error*).
3. **Configure NSX backup** (ECDSA fingerprint, relative path) → re-run (*clears the NSX backup error + schedule warning*).
4. **Silence "vSAN SCSI controller"** if still red (nested limitation).
5. **Re-run precheck** → **target ZERO errors**. Warnings may remain; they do not block.

Cautions (this lab, hard-won): - **Do NOT blind-"Apply All" the six Configuration Updates.** Several (service account, SDDC-Manager-as-extension, online-depot registration) are redeploy fallout and are fine to apply — but apply **deliberately**, verifying each against the precheck errors, given this lab's cascade history. - **Re-run**

timing: the failing precheck ran at 20:00:59, during the NSX Error-101 reboot + a fleet-wide VM resize. **Some checks can throw transient errors mid-recovery** — a clean re-run once everything settles may itself drop the count. Confirm NSX is green (System → Appliances) first. - **9.0.2 is a patch, not urgent.** Do not rush the apply on a cluster that has just been through a management-plane recovery. Stage, clear, verify — then schedule the patch deliberately. - The **cluster-store (DKVS) loop** (KB 385346) is still active but was **not** flagged as a precheck error here; it remains a watch item, not a 9.0.2 blocker. See [[RUNBOOK-ESX-Cluster-Store-DKVS-Loop-vpxd-Troubleshooting-2026-07-27]].

8. Broadcom Sources (verified 2026-07-29)

- **Perform Update Precheck** (silence/restore, filters) — [.../9-0/deployment/upgrading-cloud-foundation/.../perform-update-precheck.html](#)
- **Patching the Management and Workload Domains** — [.../9-0/lifecycle-management/lifecycle-management-of-vcf-core-components/patching-the-management-and-workload-domains.html](#)
- **vSAN SCSI controller check + OSA-only scope** — KB 400362 (points to KB 325347 for real OSA cert issues)
- **vSAN storage-controller precheck backend bypass** — **4.x/5.x ONLY** — KB 386879 (*UNVERIFIED for 9.0 — do not use here*)
- **vCenter service account discovered/disco + Remediate** — KB 399460
- **Recover expired SSO service accounts (90-day)** — KB 314177
- **Remediate passwords (9.0)** — [.../9-0/fleet-management/manage-passwords/remediate-passwords.html](#)
- **NSX backup precheck ("configured + ≥1 backup"), expired backup user** — KB 330382
- **NSX backup schedule auto-re-enable (fixed 9.0.2)** — KB 428140
- **vCenter patch-store URL / token update** — KB 393951 (and KB 390120)
- **Configure an online depot / download token** — [.../9-0/lifecycle-management/binary-management-for-vmware-re-cloud-foundation/.../configure-an-online-depot.html](#)
- **DRS Fully Automated for upgrade** — KB 426719
- **NSX Manager cert CN/SAN** — KB 419000 (regen via VMCA: KB 317900)

(All under [techdocs.broadcom.com/us/en/vmware-cis/](#) or [knowledge.broadcom.com](#). Item tags: [V] cited above · [V-LIVE] verify in your UI · [UNVERIFIED] not confirmed.)

9. Cross-References

- Offline depot config + server: [[reference_offline_depot_config_values]], [[reference_vcf_depot_server]], [[Fleet-Offline-Depot-Metadata-Mismatch-2026-05-26]]
- vCenter service account: [[reference_sddc_vcenter_service_account_recreate]], [[MODULE-SDDC-Manager-Credential-Reconnect-After-vCenter-Redeploy-2026-06-13]]
- NSX backup / SFTP: [[reference_nsx_sftp_fingerprint_format]], [[reference_sftp_backup_chroot_path]]
- Cluster-store watch: [[RUNBOOK-ESX-Cluster-Store-DKVS-Loop-vpxd-Troubleshooting-2026-07-27]]
- Upgrade context (9.0→9.1 differs): [[RUNBOOK-VCF-9.0-to-9.1-Upgrade-and-Homelab-Feasibility-2026-07-27]]

Document Information

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Revision History

Version	Date	Notes
1.0	2026-07-29	Initial remediation runbook from the live 26-error / 4-warning 9.0.2 precheck: four root causes (depot bundle, vCenter service account, NSX backup, vSAN controller) with fixes, order, gate, and lab-specific cautions.
1.1	2026-07-29	Verified every check against Broadcom TechDocs/KB; added VERIFIED/V-LIVE/UNVERIFIED tags + a §8 Broadcom sources list. Key corrections: (a) vSAN SCSI controller = HCL check per KB 400362 (OSA-only scope; silence in UI — do NOT use the 4.x/5.x KB 386879 backend bypass on 9.0); (b) Check Service Account — decide MISSING vs DRIFTED first (recreate vs Remediate; Remediate on a missing acct jams locks); (c) NSX backup — documented cause is the expired SDDC backup user (KB 330382); (d) warnings mapped to KBs (DRS 426719, cert CN 419000, backup schedule 428140).

Document End